Module I: Introduction to IoT

#	Question	Marks	Month & Year
1	Explain functional blocks of IoT	5	Dec 2023
2	Explain functional blocks of IoT	5	Dec 2024
3	Discuss the functional blocks of IoT architecture, highlighting their roles and interactions. Provide examples to illustrate the importance of each block in the overall functionality of IoT systems	10	May 2024
4	Differentiate briefly Physical design of IoT and Logical design of IoT	5	Dec 2024
5	Differentiate briefly Physical design of IoT and Logical design of IoT	5	Dec 2025
6	Explain the characteristic of IoT	5	May 2024
7	Define Smart Object in IoT and its characteristics	5	Dec 2025
8	Define Smart Object in IoT and its characteristics	5	May 2025
9	Elaborate the Smart Object with diagram and describe its characteristics	10	May 2024
10	Elaborate the Smart Object with diagram and describe all the essential components required to make the object into the smart object	10	May 2023
11	Explain data acquisition in IoT	5	May 2025

Module II: IoT Architecture

#	Question	Marks	Month & Year
1	Elaborate the need of new network architecture in IoT	10	May 2023
2	Elaborate the need of new network architecture in IoT	10	May 2024
3	Describe IoT World Forum (IoTWF) Standardized Architecture	10	May 2023
4	Describe IoT World Forum (IoTWF) standardized architecture	10	May 2025
5	With neat diagram, elaborate briefly the simplified 3 layered IoT architecture	10	Dec 2023
6	Elaborate briefly the simplified 3 layered IoT architecture with neat diagram	10	Dec 2024
7	Elaborate briefly the simplified 3 layered IoT architecture with neat diagram	10	Dec 2025
8	Define the following: 1) Fog Computing 2) Internet of Behaviour (IoB)	5	Dec 2024
9	Define the following: 1) Fog Computing 2) Internet of Behaviour (IoB)	5	Dec 2025
10	Differentiate and give Hierarchy between Edge, Fog, and Cloud computing	10	Dec 2023

#	Question	Marks	Month & Year
11	Compare and contrast Edge, Fog, and Cloud Computing. Explain their usage with an example	10	May 2023
12	Compare edge, fog and cloud computing w.r.to its hierarchy	10	May 2024

Module III: Principles of Connected Devices and Protocols in IoT

#	Question	Marks	Month & Year
1	Explain ZigBee in brief	5	Dec 2023
2	State various operating modes of Zigbee and its topologies with diagrams	10	May 2025
3	Explain applications of Bluetooth Low Energy (BLE)	5	May 2023
4	Explain Bluetooth Low Energy (BLE) role	5	May 2024
5	Write a short note on Near Field Communication	5	Dec 2024
6	Write a short note on Near Field Communication	5	Dec 2025
7	Write short note on RFID	5	May 2025
8	Explain the architecture of RFID with its major applications	10	Dec 2025
9	Explain the following access technologies with applications area of each: 1) IEEE 802.15.4 2) RFID	10	Dec 2023
10	Explain the following access technologies with applications area of each: 1) Zigbee 2) RFID 3) 6LoWPAN 4) CoAP	10	May 2023
11	Explain the following access technologies with applications area of each: 1) IEEE 802.15.4 2) Z-wave 3) LTE-A	10	May 2024
12	Illustrate the components of IEEE 802.11 architecture with advantages	10	Dec 2024
13	Illustrate the components of IEEE 802.11 architecture with advantages	10	Dec 2025
14	Explain the Need of lightweight new communications protocols for IoT	5	Dec 2023
15	Describe the communication system and protocols involved in long range communication system of IoT	10	Dec 2023

#	Question	Marks	Month & Year
16	Evaluate long-range communication systems and protocols such as LTE, LTE-A, LoRa, and LoRaWAN in the context of IoT connectivity. Discuss their suitability for different IoT use cases based on factors like coverage, data rate, power consumption, and scalability	10	May 2024
17	Explain the architecture of LoRaWAN with its major Characteristics	10	May 2023
18	Explain the architecture of LoRaWAN with its major Characteristics	10	Dec 2024
19	Explain the architecture of LoRaWAN with its major characteristics	10	May 2025

Module IV: Edge to Cloud Protocols

#	Question	Marks	Month & Year
1	Elaborate MQTT with its working in details and two advantages over COAP	10	Dec 2023
2	Draw and explain Architecture of MQTT with diagram	10	May 2023
3	Explain in brief the MQTT Architecture with Advantages	10	Dec 2024
4	Draw and explain the architecture of MQTT with diagram	10	May 2025
5	Explain in brief the MQTT Architecture with Advantages	10	Dec 2025
6	Briefly elaborate the COAP	5	May 2024
7	Explain working of Constraint Application Layer protocol (CoAP)	10	May 2025
8	Explain the following terms: 1) CoAP 2) Data Lakes	10	Dec 2024
9	Explain the following terms: 1) Constrained Application Protocol 2) Streaming Text Oriented Message Protocol	10	Dec 2025
10	Briefly elaborate the Advanced Message Queuing Protocol (AMQP)	5	May 2023
11	Describe the Architecture of Advanced Message Queuing Protocol (AMQP) with major application	10	Dec 2024
12	Describe the Architecture of Advanced Message Queuing Protocol (AMQP) with major application	10	Dec 2025
13	Draw the Architecture diagram of Advanced Message Queuing Protocol (AMQP) with advantages	5	May 2025
14	State the Comparison's of all Protocols used in IOT for cloud Edge	10	Dec 2024
15	State the Comparison's of all Protocols used in IOT for cloud Edge	10	Dec 2025

#	Question	Marks	Month & Year
16	Explain the role of HTTP, WebSocket, and MQTT in IoT communication. Compare and contrast these protocols in terms of their characteristics, suitability for different IoT scenarios, and support for real-time data transmission	10	May 2024

Module V: IoT and Data Analytics

#	Question	Marks	Month & Year
1	Explain the importance of IoT Data Analytics	5	Dec 2024
2	Explain the importance of IoT Data Analytics	5	Dec 2025
3	Explain the importance of IoT Data Analytics	5	May 2025
4	Define the role of analytics in IoT technology and elaborate the challenges associated with it	10	May 2024
5	Compare and contrast IoT data analytics and IoT network analytics	5	May 2023
6	Describe the role of NoSQL in IoT Data Analytics Challenges	10	Dec 2024
7	Describe the role of NoSQL in IoT Data Analytics Challenges	10	Dec 2025
8	Explain the role of NoSQL in IoT data analytics	10	May 2025
9	Elaborate in detail the strategies to organize Data for IoT Analytics	10	Dec 2023
10	Illustrate the role of Data refineries in preventing data lakes to turn into data swamps	10	Dec 2023
11	Explain data retention strategy	5	Dec 2023
12	Explain data retention strategy	5	May 2024
13	Illustrate 3 methods for effective Data Visualization	10	Dec 2023
14	Explain effective Data Visualization methods With suitable example	10	Dec 2024
15	Explain effective Data Visualization methods With suitable example	10	Dec 2025
16	State advantages of data visualisation. Give different tools used for it	10	May 2023
17	Illustrate the following: 1) How to create and visualize alerts in IoT? 2) What are the main challenges of IoT?	10	May 2025

#	Question	Marks	Month & Year
18	How can IoT analytics be effectively utilized within IoT-based healthcare systems? Additionally, what are some essential parameters that should be incorporated into the patient dashboard for comprehensive monitoring and management of health data?	10	May 2024

Module VI: IoT Application Design

#	Question	Marks	Month & Year
1	Define sensor, actuators and their role and classifications in brief	10	Dec 2023
2	While designing the IoT based smoke detection system what all parameters need to be considered before choosing the required sensor	5	May 2023
3	Explain the IOT enabled Home Automation system with respect to sensors, Actuators, framework, protocols, data analysis and security	10	Dec 2024
4	Explain ecosystem for IoT enabled Smart Home Automation with respect to sensors, actuators, framework, protocols, storage, data analysis, security etc	10	May 2023
5	Describe IoT enabled Smart Home Automation	10	May 2025
6	Explain the IOT enabled Smart Parking system with respect to sensors, Actuators, framework, protocols, data analysis and security	10	Dec 2025
7	Consider smart farming system. Elaborate its working and list down the different types of sensors and actuators required during the deployment scenario	10	May 2023
8	Consider Smart Irrigation system. Elaborate its working with block diagram and list down the different types of sensors and actuators required during the deployment scenario	10	Dec 2023
9	Consider smart smoke detection system. Elaborate its working and list down the different types of sensors and actuators required during the deployment scenario	10	May 2024
10	Design the Forest Fire Detection system using IoT sensors	10	May 2025
11	Write a short note on I-IoT	5	Dec 2023
12	Explain the concept of I-IoT and its similarity with IoT	5	May 2024
13	What is the Internet of Behaviour (IoB) and why is it the future?	5	May 2023
14	Explain the following terms: 1) CoAP 2) Internet of Behaviour	10	May 2025